

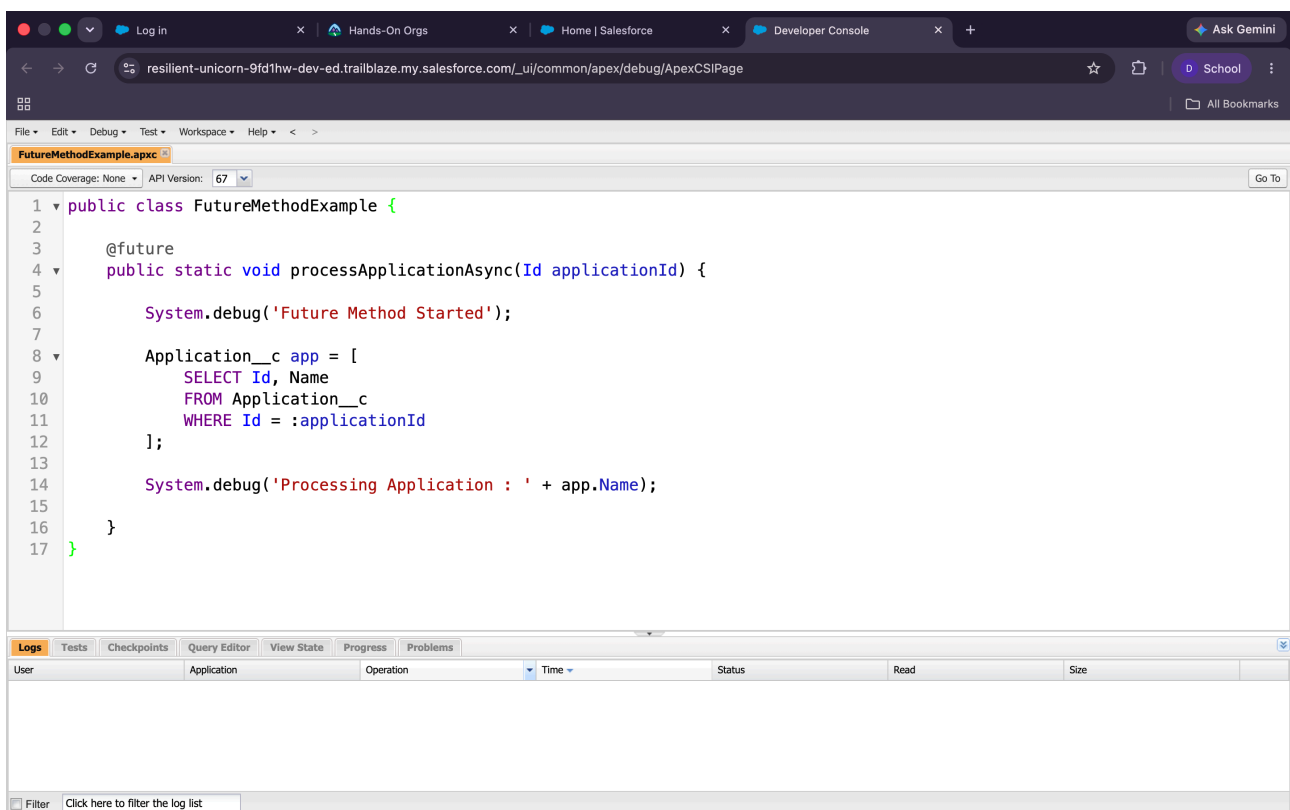
Experiment 1: Future Method Apex

Objective

To create a Future Method using @future annotation to execute background processing asynchronously without making the user wait.

Description

Future Methods are used to perform operations in the background after the current transaction is completed. They improve application performance by executing non-immediate tasks asynchronously. In this experiment, a Future Method is created to retrieve an Account record and display its details in the debug log.



Apex Code:

```
public class FutureMethodExample {

    @future
    public static void processApplicationAsync(Id accountId) {

        System.debug('Future Method Started');

        Account acc = [
            SELECT Id, Name
            FROM Account
            WHERE Id = :accountId
```

```

    ];

    System.debug('Processing Account: ' + acc.Name);
}
}
}

```

Experiment 2: Execute Future Method

Objective

To execute the Future Method using Execute Anonymous and verify that the asynchronous process starts successfully.

Description

The Future Method is executed from the Developer Console using Execute Anonymous. The Account ID is passed as an input parameter, and Salesforce schedules the method for asynchronous execution.

Execute Anonymous Code:

```
FutureMethodExample.processApplicationAsync('001hg000007UBBhAAO');
```

The screenshot displays the Salesforce Developer Console interface. The top navigation bar shows the current context: 'resilient-unicorn-9fd1hw-dev-ed.trailblaze.my.salesforce.com/_ui/common/apex/debug/ApexCSIPage'. The main area is divided into two sections: 'Execution Log' and 'Logs'.

Execution Log: This section shows a detailed log of the execution process. The log entries include timestamps, event types, and details. Key events include 'USER_INFO', 'EXECUTION_START', 'CODE_UNIT_START', 'CODE_UNIT_FINISH', 'EXECUTION_FINISH', 'HEAP_ALLOCATE', 'METHOD_ENTRY', and 'STATEMENT_EXECUTE'. The log indicates that the 'FutureMethodExample.processApplicationAsync' method was executed successfully.

Logs: This section provides a summary of the execution logs. It includes columns for 'User', 'Application', 'Operation', 'Time', 'Status', 'Read', and 'Size'. The logs show that the 'FutureMethodExample.processApplicationAsync' method was executed successfully on 07/08/2026 at 10:28:22.

User	Application	Operation	Time	Status	Read	Size
DUDALA MADHAVI	Unknown	/services/data/v67.0/tooling/executeAnonymous	07/08/2026, 10:28:22	Success	Unread	2.23 KB
DUDALA MADHAVI	Unknown	FutureHandler	07/08/2026, 10:28:22	Success	Unread	4.06 KB
DUDALA MADHAVI	Unknown	/services/data/v67.0/tooling/executeAnonymous	07/08/2026, 10:14:49	Success	Unread	2.23 KB
DUDALA MADHAVI	Unknown	FutureHandler	07/08/2026, 10:14:49	List has no rows for assignment to ...		4 KB

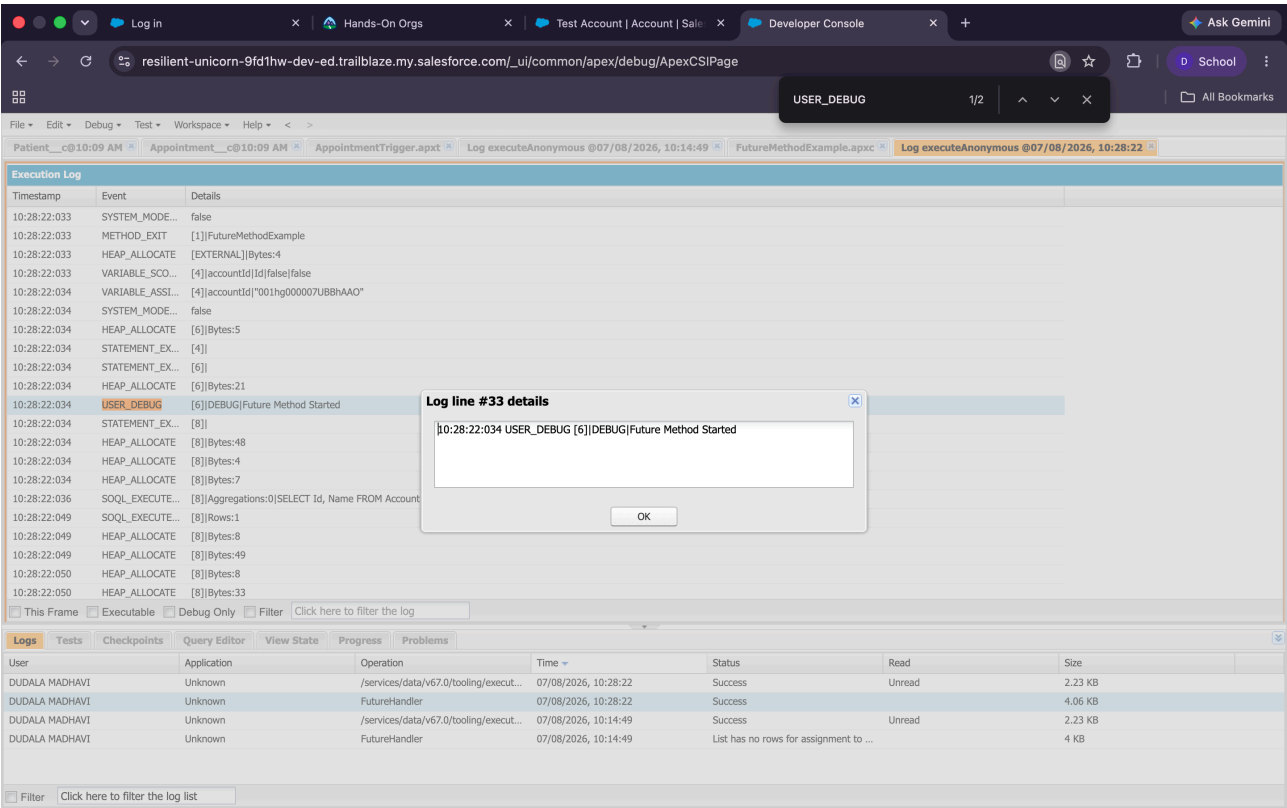
Experiment 3: Debug Log Verification

Objective

To verify that the Future Method has started successfully using the Salesforce Debug Log.

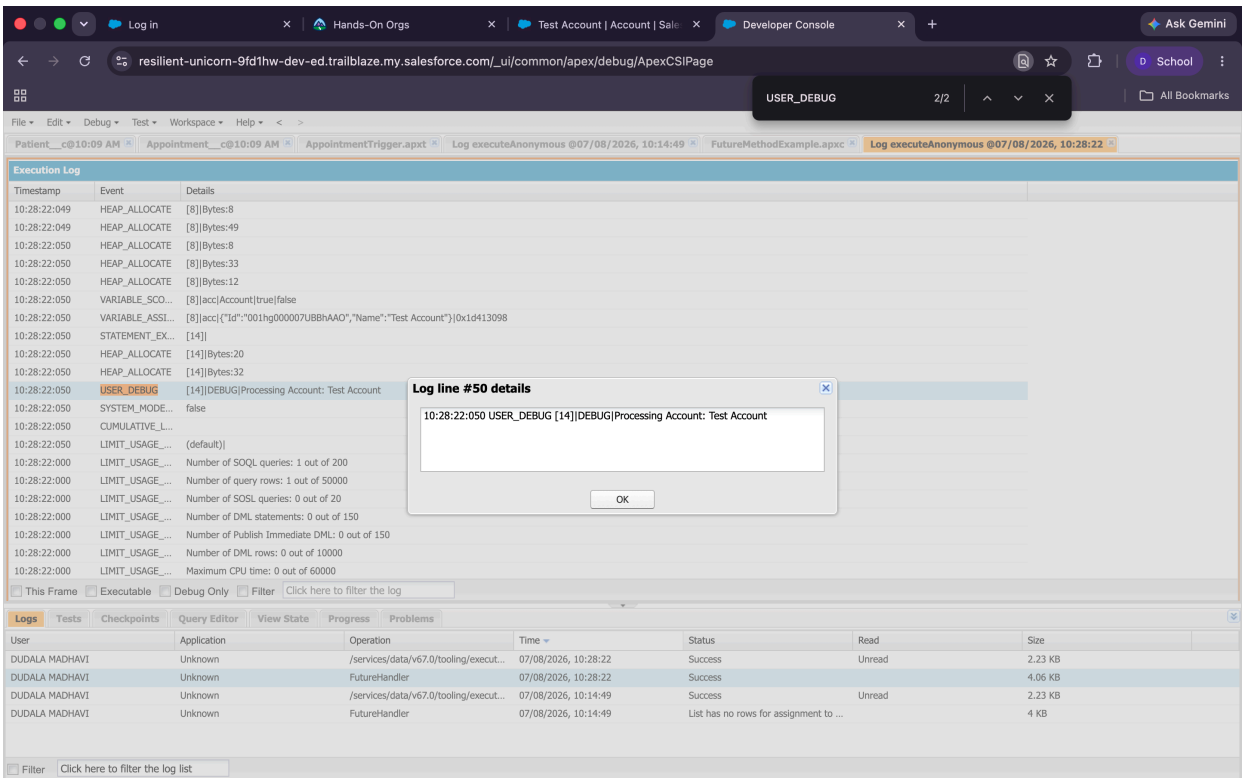
Description

The Debug Log displays the message **"Future Method Started"**, confirming that Salesforce executed the Future Method asynchronously.



Experiment 4: Processing Account

Objective



To verify that the Future Method successfully retrieved and processed the Account record.

Description

The Debug Log displays the Account name that was processed by the Future Method, confirming successful execution.

Experiment 5: Queueable Apex

Objective

To create a Queueable Apex class for structured background processing.

Description

Queueable Apex allows developers to perform asynchronous processing with better control, scalability, and flexibility. The Queueable class processes Account records and prints debug messages.

Apex Code

```
public class AccountQueueableJob implements Queueable {  
  
    public void execute(QueueableContext context) {  
  
        System.debug('Queueable Job Started');  
  
        List<Account> accList = [  
            SELECT Id, Name  
            FROM Account  
        ];  
  
        for(Account acc : accList){  
            System.debug('Processing Account: ' + acc.Name);  
        }  
  
        System.debug('Queueable Job Completed');  
    }  
}
```

The screenshot displays the Salesforce Developer Console interface. The top section shows the Apex code for the `AccountQueueableJob` class, which implements the `Queueable` interface. The code includes a `execute` method that logs the start of the job, queries for all Account records, iterates through them logging each name, and logs the completion of the job. The bottom section shows the 'Logs' tab, which contains a table of debug logs for the user 'DUDALA MADHAVI'. The logs show three entries: two successful executions of the `FutureHandler` and one unread log entry.

User	Application	Operation	Time	Status	Read	Size
DUDALA MADHAVI	Unknown	/services/data/v67.0/tooling/execut...	07/08/2026, 10:28:22	Success	Unread	2.23 KB
DUDALA MADHAVI	Unknown	FutureHandler	07/08/2026, 10:28:22	Success		4.06 KB
DUDALA MADHAVI	Unknown	/services/data/v67.0/tooling/execut...	07/08/2026, 10:14:49	Success	Unread	2.23 KB
DUDALA MADHAVI	Unknown	FutureHandler	07/08/2026, 10:14:49	List has no rows for assignment to ...		4 KB

```

        for(Account acc : accList){
            System.debug('Processing Account: ' + acc.Name);
        }

        System.debug('Queueable Job Completed');
    }
}

```

Experiment 6: Execute Queueable Apex

Objective

To execute the Queueable Apex job asynchronously.

Execute Anonymous Code

```
System.enqueueJob(new AccountQueueableJob());
```

The screenshot shows the Salesforce Developer Console interface. The top bar indicates the user is logged in and the current context is 'Test Account | Account | Sales'. The browser address bar shows the URL: `resilient-unicorn-9fd1hw-dev-ed.trailblaze.my.salesforce.com/_ui/common/apex/debug/ApexCSIPage`. The console title is 'AccountQueueableJob.apxc' and it shows a log entry for 'Log executeAnonymous @07/08/2026, 11:19:51'.

The 'Execution Log' tab is active, displaying a table of events:

Timestamp	Event	Details
11:19:51:001	USER_INFO	[EXTERNAL] 005hg000001en1R 23pa1a1258@resilient-unicorn-9fd1hw.com (GMT+05:30) India Standard Time (Asia/Kolkata) GMT+05:30
11:19:51:001	EXECUTION_ST...	
11:19:51:001	CODE_UNIT_ST...	[EXTERNAL] execute_anonymous_apex
11:19:51:002	HEAP_ALLOCATE	[95] Bytes:3
11:19:51:002	HEAP_ALLOCATE	[100] Bytes:152
11:19:51:002	HEAP_ALLOCATE	[417] Bytes:408
11:19:51:002	HEAP_ALLOCATE	[430] Bytes:408
11:19:51:002	HEAP_ALLOCATE	[317] Bytes:6
11:19:51:002	HEAP_ALLOCATE	[EXTERNAL] Bytes:1
11:19:51:002	STATEMENT_EX...	[1]
11:19:51:002	STATEMENT_EX...	[1]
11:19:51:010	HEAP_ALLOCATE	[1] Bytes:1
11:19:51:010	HEAP_ALLOCATE	[1] Bytes:7
11:19:51:011	METHOD_ENTRY	[1] 01.phg000001paF8 AccountQueueableJob.AccountQueueableJob()
11:19:51:011	STATEMENT_EX...	[1]
11:19:51:011	STATEMENT_EX...	[1]
11:19:51:011	HEAP_ALLOCATE	[68] Bytes:5
11:19:51:011	HEAP_ALLOCATE	[74] Bytes:5
11:19:51:011	HEAP_ALLOCATE	[82] Bytes:7
11:19:51:011	SYSTEM_MODE...	false
11:19:51:011	HEAP_ALLOCATE	[1] Bytes:5

Below the execution log, there are tabs for 'Logs', 'Tests', 'Checkpoints', 'Query Editor', 'View State', 'Progress', and 'Problems'. The 'Logs' tab is active, showing a table of log entries:

User	Application	Operation	Time	Status	Read	Size
DUDALA MADHAVI	Unknown	/services/data/v67.0/tooling/execut...	07/08/2026, 11:19:51	Success		2.95 KB
DUDALA MADHAVI	Unknown	QueueableHandler	07/08/2026, 11:19:51	Success	Unread	10.8 KB
DUDALA MADHAVI	Unknown	/services/data/v67.0/tooling/execut...	07/08/2026, 10:28:22	Success	Unread	2.23 KB
DUDALA MADHAVI	Unknown	FutureHandler	07/08/2026, 10:28:22	Success		4.06 KB
DUDALA MADHAVI	Unknown	/services/data/v67.0/tooling/execut...	07/08/2026, 10:14:49	Success	Unread	2.23 KB
DUDALA MADHAVI	Unknown	FutureHandler	07/08/2026, 10:14:49	List has no rows for assignment to ...		4 KB

Experiment 7: Queueable Debug Messages

Objective

To verify successful execution of the Queueable Apex job using Debug Logs.

Description

The Debug Log displays:

Execution Log

Timestamp	Event	Details
11:21:17:020	VARIABLE_ASSI...	[12] acc {"Id":"001hg0000068IVYAA0","Name":"Sample Account for E (11 more) ..."} 0x7b8126d0
11:21:17:020	STATEMENT_EX...	[12]
11:21:17:020	STATEMENT_EX...	[13]
11:21:17:020	HEAP_ALLOCATE	[13] Bytes:51
11:21:17:020	USER_DEBUG	[13] DEBUG Processing Account: Sample Account for Entitlements
11:21:17:020	HEAP_ALLOCATE	[12] Bytes:5
11:21:17:020	HEAP_ALLOCATE	[12] Bytes:12
11:21:17:020	VARIABLE_SCO...	[12] acc Account true false
11:21:17:020	VARIABLE_ASSI...	[12] acc {"Id":"001hg000007UBBhAA0","Name":"Test Account"} 0x532f286
11:21:17:020	STATEMENT_EX...	[12]
11:21:17:020	STATEMENT_EX...	[13]
11:21:17:020	HEAP_ALLOCATE	[13] Bytes:32
11:21:17:020	USER_DEBUG	[13] DEBUG Processing Account: Test Account
11:21:17:020	HEAP_ALLOCATE	[12] Bytes:5
11:21:17:020	VARIABLE_ASSI...	[12] acc null
11:21:17:020	STATEMENT_EX...	[16]
11:21:17:020	HEAP_ALLOCATE	[16] Bytes:23
11:21:17:020	USER_DEBUG	[16] DEBUG Queueable Job Completed
11:21:17:020	SYSTEM_MODE...	false
11:21:17:020	CODE_UNIT_FL...	AccountQueueableJob
11:21:17:020	EXECUTION_FL...	

Log line #169 details

11:21:17:020 USER_DEBUG [16]]DEBUG|Queueable Job Completed

OK

User	Application	Operation	Time	Status	Read	Size
DUDALA MADHAVI	Unknown	QueueableHandler	07/08/2026, 11:21:17	Success		10.82 KB
DUDALA MADHAVI	Unknown	/services/data/v67.0/tooling/execut...	07/08/2026, 11:21:16	Success	Unread	2.93 KB
DUDALA MADHAVI	Unknown	/services/data/v67.0/tooling/execut...	07/08/2026, 11:19:51	Success		2.95 KB
DUDALA MADHAVI	Unknown	QueueableHandler	07/08/2026, 11:19:51	Success	Unread	10.8 KB
DUDALA MADHAVI	Unknown	/services/data/v67.0/tooling/execut...	07/08/2026, 10:28:22	Success	Unread	2.23 KB
DUDALA MADHAVI	Unknown	FutureHandler	07/08/2026, 10:28:22	Success		4.06 KB

- Queueable Job Started
- Processing Account
- Queueable Job Completed

These messages confirm successful background execution.

Experiment 8: Batch Apex

```
1 public class AccountBatchJob implements Database.Batchable<SObject> {
2
3     public Database.QueryLocator start(Database.BatchableContext bc) {
4
5         return Database.getQueryLocator(
6             'SELECT Id, Name FROM Account'
7         );
8     }
9
10    public void execute(Database.BatchableContext bc, List<Account> scope) {
11
12        for(Account acc : scope){
13            System.debug('Batch Processing: ' + acc.Name);
14        }
15    }
16
17    public void finish(Database.BatchableContext bc){
18
19        System.debug('Batch Job Completed');
20    }
```

User	Application	Operation	Time	Status	Read	Size
DUDALA MADHAVI	Unknown	QueueableHandler	07/08/2026, 11:21:17	Success		10.82 KB
DUDALA MADHAVI	Unknown	/services/data/v67.0/tooling/execut...	07/08/2026, 11:21:16	Success	Unread	2.93 KB
DUDALA MADHAVI	Unknown	/services/data/v67.0/tooling/execut...	07/08/2026, 11:19:51	Success		2.95 KB
DUDALA MADHAVI	Unknown	QueueableHandler	07/08/2026, 11:19:51	Success	Unread	10.8 KB
DUDALA MADHAVI	Unknown	/services/data/v67.0/tooling/execut...	07/08/2026, 10:28:22	Success	Unread	2.23 KB
DUDALA MADHAVI	Unknown	FutureHandler	07/08/2026, 10:28:22	Success		4.06 KB

Objective

To create a Batch Apex class that processes Account records in batches.

Description

Batch Apex is used to process a large number of records efficiently by dividing them into smaller batches. The Batch class includes the start(), execute(), and finish() methods.

Experiment 9: Batch Execution

Objective

To execute the Batch Apex class using Execute Anonymous.

Execute Anonymous Code

Database.executeBatch(new AccountBatchJob(), 5);

Log inHands-On OrgsTest Account | Account | SalesDeveloper ConsoleAsk Gemini

resilient-unicorn-9fd1hw-dev-ed.trailblaze.my.salesforce.com/_ui/common/apex/debug/ApexCSIPage

AccountBatchJob.apxcLog executeAnonymous @07/08/2026, 11:24:49

Execution Log

Timestamp	Event	Details
11:24:49:000	USER_INFO	[EXTERNAL] 005hg000001en1R 23pa1a1258@resilient-unicorn-9fd1hw.com (GMT+05:30) India Standard Time (Asia/Kolkata) (GMT+05:30)
11:24:49:000	EXECUTION_ST...	
11:24:49:000	CODE_UNIT_ST...	[EXTERNAL] 01phg000001pbY1 AccountBatchJob
11:24:49:003	HEAP_ALLOCATE	[EXTERNAL] Bytes:4
11:24:49:015	HEAP_ALLOCATE	[95] Bytes:3
11:24:49:015	HEAP_ALLOCATE	[100] Bytes:152
11:24:49:015	HEAP_ALLOCATE	[417] Bytes:408
11:24:49:015	HEAP_ALLOCATE	[430] Bytes:408
11:24:49:015	HEAP_ALLOCATE	[317] Bytes:6
11:24:49:015	HEAP_ALLOCATE	[EXTERNAL] Bytes:11
11:24:49:015	SYSTEM_METH...	[15] BatchableContextImpl.BatchableContextImpl()
11:24:49:015	STATEMENT_EX...	[15]
11:24:49:015	SYSTEM_METH...	[15] BatchableContextImpl
11:24:49:015	HEAP_ALLOCATE	[EXTERNAL] Bytes:12
11:24:49:015	HEAP_ALLOCATE	[EXTERNAL] Bytes:4
11:24:49:015	VARIABLE_SCO...	[31] this Database.BatchableContextImpl true false
11:24:49:015	VARIABLE_ASSI...	[31] this {} 0x41776965
11:24:49:015	VARIABLE_SCO...	[31] jobId Id false false
11:24:49:015	VARIABLE_ASSI...	[31] jobId "707hg000001jy5AAE"
11:24:49:015	VARIABLE_SCO...	[31] childJobId Id false false
11:24:49:015	VARIABLE_ASSI...	[31] childJobId null

☐ This Frame☐ Executable☐ Debug Only☐ FilterClick here to filter the log

LogsTestsCheckpointsQuery EditorView StateProgressProblems

User	Application	Operation	Time	Status	Read	Size
DUDALA MADHAVI	Unknown	Batch Apex	07/08/2026, 11:24:50	Success	Unread	3.7 KB
DUDALA MADHAVI	Unknown	SerialBatchApexRangeChunkHandler	07/08/2026, 11:24:50	Success	Unread	5.52 KB
DUDALA MADHAVI	Unknown	SerialBatchApexRangeChunkHandler	07/08/2026, 11:24:50	Success	Unread	4.78 KB
DUDALA MADHAVI	Unknown	/services/data/v67.0/tooling/execut...	07/08/2026, 11:24:49	Success	Unread	2.98 KB
DUDALA MADHAVI	Unknown	Batch Apex	07/08/2026, 11:24:49	Success		3.83 KB
DUDALA MADHAVI	Unknown	SerialBatchApexRangeChunkHandler	07/08/2026, 11:24:49	Success	Unread	2.53 KB

☐ FilterClick here to filter the log list

Experiment 10: Batch Debug Messages

Objective

To verify successful execution of the Batch Apex job using Debug Logs.

Description

resilient-unicorn-9fd1hw-dev-ed.trailblaze.my.salesforce.com/_ui/common/apex/debug/ApexCSIPage

USER_DEBUG 1/1

Log executeAnonymous @07/08/2026, 11:26:56

Execution Log

Timestamp	Event	Details
11:26:56:022	System.out.println	[18]
11:26:56:027	STATEMENT_EXECUTE	[20]
11:26:56:027	HEAP_ALLOCATE	[20]Bytes:19
11:26:56:027	USER_DEBUG	[20]DEBUG Batch Job Completed
11:26:56:027	SYSTEM_MODE...	false
11:26:56:051	CUMULATIVE_L...	
11:26:56:051	LIMIT_USAGE...	(default)
11:26:56:000	LIMIT_USAGE...	Number of SOQL queries: 0 out of 200
11:26:56:000	LIMIT_USAGE...	Number of query rows: 0 out of 50000
11:26:56:000	LIMIT_USAGE...	Number of Sosl queries: 0 out of 20
11:26:56:000	LIMIT_USAGE...	Number of DML statements: 0 out of 150
11:26:56:000	LIMIT_USAGE...	Number of Publish Immediate DML: 0 out of 150
11:26:56:000	LIMIT_USAGE...	Number of DML rows: 0 out of 10000
11:26:56:000	LIMIT_USAGE...	Maximum CPU time: 0 out of 60000
11:26:56:000	LIMIT_USAGE...	Maximum heap size: 0 out of 12000000
11:26:56:000	LIMIT_USAGE...	Number of callouts: 0 out of 0
11:26:56:000	LIMIT_USAGE...	Number of Email Invocations: 0 out of 10
11:26:56:000	LIMIT_USAGE...	Number of future calls: 0 out of 0
11:26:56:000	LIMIT_USAGE...	Number of queueable jobs added to the queue: 0 out of 1
11:26:56:000	LIMIT_USAGE...	Number of Mobile Apex push calls: 0 out of 10
11:26:56:000	LIMIT_USAGE...	

Log line #45 details

11:26:56:027 USER_DEBUG [20]DEBUG|Batch Job Completed

OK

Logs Tests Checkpoints Query Editor View State Progress Problems

User	Application	Operation	Time	Status	Read	Size
DUDALA MADHAVI	Unknown	/services/data/v67.0/tooling/execut...	07/08/2026, 11:26:56	Success	Unread	2.95 KB
DUDALA MADHAVI	Unknown	Batch Apex	07/08/2026, 11:26:56	Success	Unread	3.7 KB
DUDALA MADHAVI	Unknown	Batch Apex	07/08/2026, 11:26:56	Success	Unread	3.83 KB
DUDALA MADHAVI	Unknown	SerialBatchApexRangeChunkHandler	07/08/2026, 11:26:56	Success	Unread	2.53 KB
DUDALA MADHAVI	Unknown	SerialBatchApexRangeChunkHandler	07/08/2026, 11:26:56	Success	Unread	4.77 KB
DUDALA MADHAVI	Unknown	SerialBatchApexRangeChunkHandler	07/08/2026, 11:26:56	Success	Unread	5.51 KB

Filter Click here to filter the log list

The Debug Log displays **"Batch Job Completed"**, confirming successful batch execution.

Experiment 11: Scheduled Apex

Objective

To create a Scheduled Apex class that executes automatically at a specified time.

Description

Scheduled Apex automates background processing based on a defined schedule using the Schedulable interface

resilient-unicorn-9fd1hw-dev-ed.trailblaze.my.salesforce.com/_ui/common/apex/debug/ApexCSIPage

USER_DEBUG 1/1

AccountScheduler.apxc

Code Coverage: None API Version: 67

```
1 public class AccountScheduler implements Schedulable {
2
3     public void execute(SchedulableContext sc) {
4
5         System.debug('Scheduled Job Started');
6
7         List<Account> accounts = [
8             SELECT Id, Name
9             FROM Account
10        ];
11
12        for(Account acc : accounts){
13            System.debug('Scheduled Processing: ' + acc.Name);
14        }
15
16        System.debug('Scheduled Job Completed');
17    }
18 }
```

Logs Tests Checkpoints Query Editor View State Progress Problems

User	Application	Operation	Time	Status	Read	Size
DUDALA MADHAVI	Unknown	/services/data/v67.0/tooling/execut...	07/08/2026, 11:26:56	Success	Unread	2.95 KB
DUDALA MADHAVI	Unknown	Batch Apex	07/08/2026, 11:26:56	Success	Unread	3.7 KB
DUDALA MADHAVI	Unknown	Batch Apex	07/08/2026, 11:26:56	Success	Unread	3.83 KB
DUDALA MADHAVI	Unknown	SerialBatchApexRangeChunkHandler	07/08/2026, 11:26:56	Success	Unread	2.53 KB
DUDALA MADHAVI	Unknown	SerialBatchApexRangeChunkHandler	07/08/2026, 11:26:56	Success	Unread	4.77 KB
DUDALA MADHAVI	Unknown	SerialBatchApexRangeChunkHandler	07/08/2026, 11:26:56	Success	Unread	5.51 KB

Filter Click here to filter the log list

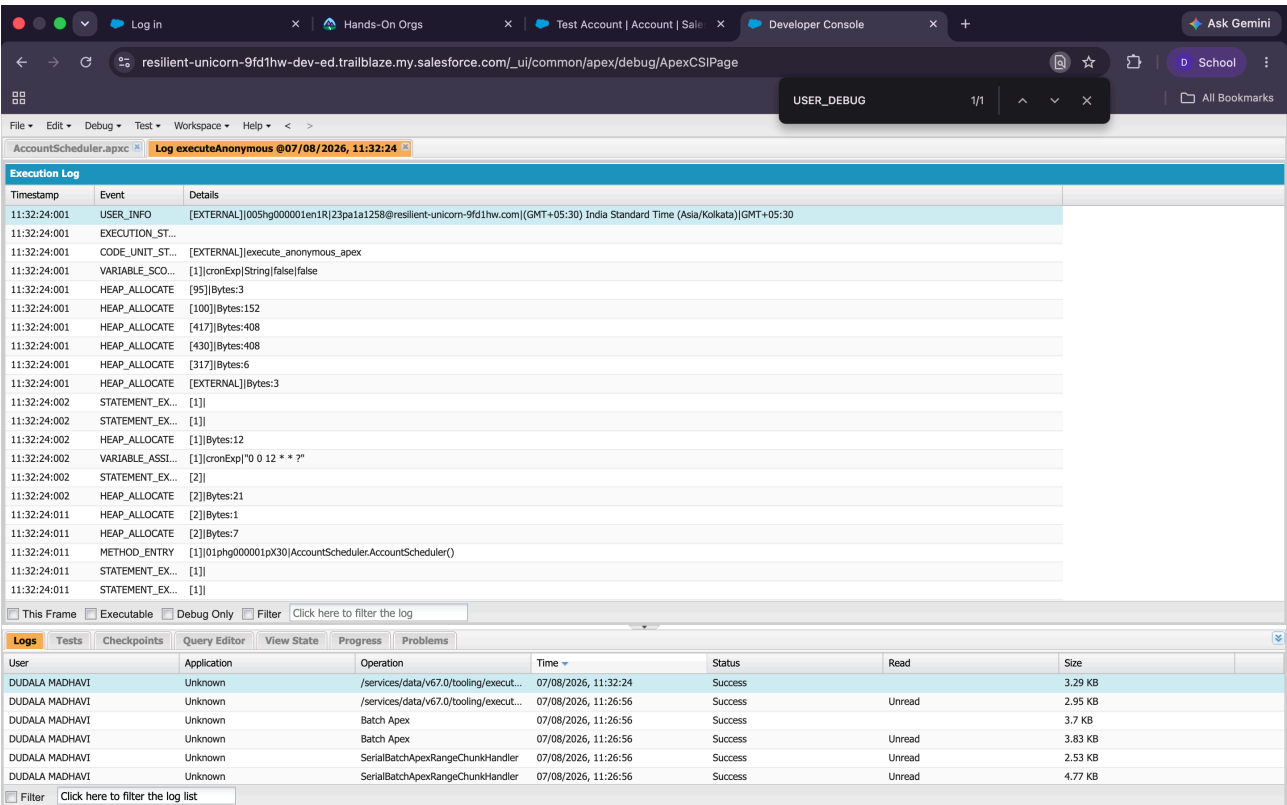
Experiment 12: Schedule Apex Job

Objective

To schedule the Apex class using a CRON expression.

Execute Anonymous Code

```
String cronExp = '0 0 12 * * ?';
System.schedule('Account Scheduler Job', cronExp, new AccountScheduler());
```



Experiment 13: Verify Scheduled Job

Objective

To verify that the Scheduled Apex job has been successfully created.

Description

The Scheduled Jobs page displays the scheduled job name, next execution time, and job type, confirming successful scheduling.

